MSG No. 138.520 $P4_2/ncm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.221'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,6,7,8,17,22,23,24]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,4,5,18,19,20,21]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,14,15,16,25,30,31,32]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,11,12,13,26,27,28,29]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6,10,11,17,22,26,27]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,14,18,19,25,30]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,5,15,16,20,21,31,32]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,8,12,13,23,24,28,29]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2/m1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,8,9,16,17,24,25,32]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,4,10,12,18,20,26,28]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,5,11,13,19,21,27,29]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,7,14,15,22,23,30,31]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2/m1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,8,9,16,17,24,25,32]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,4,10,12,18,20,26,28]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,5,11,13,19,21,27,29]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,14,15,22,23,30,31]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm1'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,6,15,16,17,22,31,32]`
2	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[2,3,12,13,18,19,28,29]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[4,5,10,11,20,21,26,27]`
4	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[7,8,9,14,23,24,25,30]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,6,17,22]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,3,18,19]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[4,5,20,21]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[7,8,23,24]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[9,14,25,30]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,11,26,27]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[12,13,28,29]`
8	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[15,16,31,32]`

Table 7: Wyckoff site: `8g`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[1,8,17,24]`
2	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[2,4,18,20]`
3	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[3,5,19,21]`
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[6,7,22,23]`
5	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[9,16,25,32]`
6	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[10,12,26,28]`
7	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[11,13,27,29]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,15,30,31]`

Table 8: Wyckoff site: `8h`, site symmetry: `..21'`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,8,17,24]`
2	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[2,4,18,20]`
3	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,5,19,21]`
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[6,7,22,23]`
5	$[-x,\ x,\ 0]$	`[9,16,25,32]`
6	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[10,12,26,28]`
7	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[11,13,27,29]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[14,15,30,31]`

Table 9: Wyckoff site: `8i`, site symmetry: `..m1'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,16,17,32]`
2	$\left[\frac{1}{2}-x,\ x,\ z+\frac{1}{2}\right]$	`[2,12,18,28]`
3	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3,13,19,29]`
4	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[4,10,20,26]`
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,11,21,27]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[6,15,22,31]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	`[7,14,23,30]`

continued ...

Table 9

No.	position	mapping
8	$[-x,\, -x,\, -z]$	`[8,9,24,25]`

Table 10: Wyckoff site: `16j`, site symmetry: `11'`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1,17]`
2	$\left[\frac{1}{2}-y,\, x,\, z+\frac{1}{2}\right]$	`[2,18]`
3	$\left[y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[3,19]`
4	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	`[4,20]`
5	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[5,21]`
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[6,22]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	`[7,23]`
8	$[-y,\, -x,\, -z]$	`[8,24]`
9	$[-x,\, -y,\, -z]$	`[9,25]`
10	$\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$	`[10,26]`
11	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[11,27]`
12	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	`[12,28]`
13	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[13,29]`
14	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	`[14,30]`
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	`[15,31]`
16	$[y,\, x,\, z]$	`[16,32]`